

Technical Audit of the Adult Census Income Algorithmic Decision-Support System

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1 Background

The purpose of the chosen Algorithmic Decision-Support System (ADS) is to perform binary classification to predict whether an individual’s annual income exceeds \$50K. This prediction is based on socio-demographic and employment features such as workclass, education, and race (drawn from the Adult Census Income dataset).

The \$50K threshold is not chosen by us. It is the predefined target label in the Adult Census Income dataset, which frames the task as a binary classification problem: whether an individual’s annual income is greater than \$50K or not. Therefore, in this audit, \$50K should not be interpreted as a universal definition of being “rich.” It is a historical cutoff from the 1994 Census-based dataset that separates the dataset’s higher-income category from its lower-income category.

This prediction matters because the income category is often used as a proxy for socio-economic status. In settings such as public policy, social science research, resource allocation, marketing, or other decision-support systems, income predictions can shape how populations are analyzed and how services are targeted. The prediction is not neutral. A wrong prediction can misrepresent a person’s economic position, and systematic errors can cause some groups to be underestimated more often than others.

For this reason, our audit does not only ask whether the model is accurate overall. We also examine whose income status the model predicts correctly, whose status it misclassifies, and whether those mistakes are concentrated in socially meaningful subgroups. This framing is important because a model can have strong aggregate accuracy while still producing unequal errors across groups.

2 Input and Output

The Adult Census Income dataset is from UCI Machine Learning Repository. These data were extracted from the 1994 Census Bureau database by Ronny Kohavi and Barry Becker (Data Mining and Visualization, Silicon Graphics). A set of reasonably clean records was extracted using the following conditions: older than 16 (AAGE), Adjusted Gross Income (AGI) greater than \$100, a strictly positive final demographic weight (AFNLWGT), and reported working more than 0 hours per week (HRSWK).

2.1 Input

The ADS utilizes 14 input features: Age (numerical), Workclass (categorical), fnlwgt (numerical), Education (categorical), Education-num (numerical), Marital-status (categorical), Occupation (categorical), Relationship (categorical), Race (categorical), Sex (categorical), Capital-gain (numerical), Capital-loss (numerical), Hours-per-week (numerical), and Native-country (categorical).

2.1.1 Numerical Features

Distributions: Features like Age and Hours-per-week follow relatively normal distributions (with a large spike at 40 hours for the latter). However, Capital-gain and Capital-loss are extremely right-skewed and zero-inflated (over 90% of individuals report zero), though non-zero values are strong predictors of high income. fnlwgt (final weight) is a continuous demographic sampling weight; logically, it lacks a causal relationship with individual income and acts as statistical noise.

Missing values: Standard numerical features contain no missing values.

	count	mean	std	min	25%	50%	75%	max
age	32561.0	38.581647	13.640433	17.0	28.0	37.0	48.0	90.0
fnlwgt	32561.0	189778.366512	105549.977697	12285.0	117827.0	178356.0	237051.0	1484705.0
education.num	32561.0	10.080679	2.572720	1.0	9.0	10.0	12.0	16.0
capital.gain	32561.0	1077.648844	7385.292085	0.0	0.0	0.0	0.0	99999.0
capital.loss	32561.0	87.303830	402.960219	0.0	0.0	0.0	0.0	4356.0
hours.per.week	32561.0	40.437456	12.347429	1.0	40.0	40.0	45.0	99.0

Table 1: Summary statistics of numerical input variables

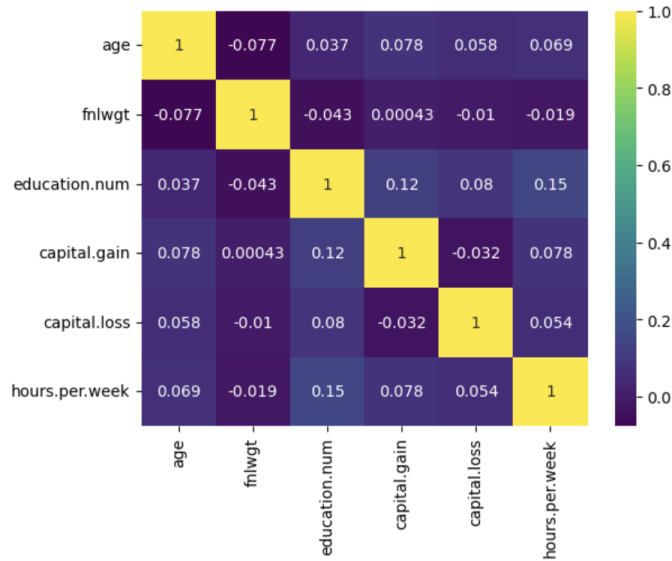


Figure 1: Correlation of numerical input variables

2.1.2 Categorical Features

Distributions: Most categorical features exhibit severe class imbalance, reflecting the socio-economic structure of 1994. For instance, race is predominantly White ($\sim 85\%$), sex is skewed toward Male ($\sim 67\%$), and native country is overwhelmingly United-States ($\sim 90\%$).

Missing values: While standard functions like `df.info()` report zero null values, missing values are encoded as the string “?”. Specifically, Workclass (1,836 missing), Occupation (1,843 missing), and Native-country (583 missing) are affected. The original author’s decision to impute these with the mode (“Private” and “Prof-specialty”) is statistically dangerous; it artificially inflates the majority class and erases the realities of marginalized individuals who might be unemployed or in informal labor. During data preprocessing, the author replaced the unknown values “?” in the “workclass” and “occupation” columns with the mode, which introduces bias into the data.

Feature	Unique	“?” Count	“?” %	Most common value (%)
workclass	9	1836	5.64	Private (69.70)
education	16	0	0.00	HS-grad (32.25)
marital.status	7	0	0.00	Married-civ-spouse (45.99)
occupation	15	1843	5.66	Prof-specialty (12.71)
relationship	6	0	0.00	Husband (40.52)
race	5	0	0.00	White (85.43)
sex	2	0	0.00	Male (66.92)
native.country	42	583	1.79	United-States (89.59)

Table 2: Categorical features

2.2 Output

The final output of this ADS is a binary class label representing an individual’s income bracket: either $> 50K$ or $\leq 50K$. Internally, the classification model (Logistic Regression) calculates a predicted probability score between 0 and 1, which is then mapped to these discrete labels using a default classification threshold (typically 0.5).

As seen in the count plot, the target class is highly imbalanced: 24720 people have income less than or equal to 50K (75.92%) and 7841 people have income greater than 50K (24.08%). This may lead to disparities in the False Negative Rate (FNR) and False Positive Rate (FPR) of the predictions.

3 Implementation and Validation

3.1 Problems in Data Preprocessing

Several preprocessing choices in the original implementation are important from a responsible data science perspective.

- The original implementation removes the **race** variable under the assumption that excluding sensitive attributes helps ensure fairness. This reflects the flawed notion of “fairness through unawareness”: removing a sensitive attribute does not guarantee fair outcomes, because other variables may still act as proxies for it.
- The original implementation replaces hidden missing values, encoded as “?”, in categorical features such as **workclass** and **occupation** with their modal values. This may hide meaningful missingness and shift uncertain cases into majority categories.
- The original implementation removes “extreme” age values, effectively excluding older individuals from the dataset. This introduces a potential sample-selection issue by narrowing the population represented in the training and evaluation data.

3.2 Implementation

After preprocessing the data, the author trains a Logistic Regression model to predict whether an individual’s income is $> 50K$ or $\leq 50K$. The implementation also uses randomized cross-validation search for hyperparameter tuning. The model outputs a binary predicted class label, although Logistic Regression can also be interpreted through its predicted probabilities and learned coefficients.

3.3 Validation

The original author evaluated the model primarily using global accuracy, reporting a seemingly robust performance of approximately 85%. However, evaluating an Algorithmic Decision-Support System (ADS) solely on aggregate accuracy is fundamentally flawed when dealing with a highly imbalanced dataset.

The author’s own global confusion matrix reveals a critical issue: the number of False Negatives (FN = 664) is substantially higher than the number of False Positives (FP = 321). This discrepancy suggests that the model is actively biasing its predictions toward the majority class ($\leq 50K$) to artificially inflate its overall accuracy.

4 Outcomes

4.1 Subgroup Performance Analysis

After reproducing the baseline Logistic Regression model, we evaluated its performance across several subpopulations, including sex, race, workclass, age group, marital status, and education. Because the target class is imbalanced, we focus especially on recall and false negative rate (FNR), rather than accuracy alone. In this task, a false negative means that an individual who actually earns $> 50K$ is predicted as earning

$\leq 50K$. This error matters because it underestimates a person’s economic status and may distort downstream analyses or resource-allocation decisions.

The sex-based results show why aggregate accuracy can be misleading. The model has higher accuracy for women than for men, 92.0% versus 81.7%. However, women also have lower recall, 50.2% compared with 60.5% for men, and a higher FNR, 49.8% compared with 39.5%. This indicates that among individuals who actually earn $> 50K$, women are more likely to be missed by the model.

Race-based performance is also uneven. Recall is 59.6% for White individuals, 51.6% for Black individuals, 56.3% for Asian-Pac-Islander individuals, and 42.9% for the Other race category. The corresponding FNRs are 40.4%, 48.4%, 43.8%, and 57.1%. These results should be interpreted carefully for smaller racial groups, but they still show that racial disparities remain even though the original implementation removes race from the training features.

Education produces the largest subgroup differences. Graduate-degree holders have a selection rate of 69.6%, recall of 90.3%, and FNR of only 9.7%. In contrast, the Less-than-HS group has a selection rate of 1.0%, recall of 15.7%, and FNR of 84.3%; the HS-grad group also has low recall, 27.5%, and high FNR, 72.5%. These gaps suggest that the model strongly relies on formal education and often fails to identify high-income individuals who do not match the typical high-education profile.

Overall, subgroup analysis shows that the baseline model does not perform equally across the population. Groups with lower positive base rates may appear to have high accuracy, but they often have much lower recall and higher false negative rates. This motivates the formal fairness analysis in the next section.

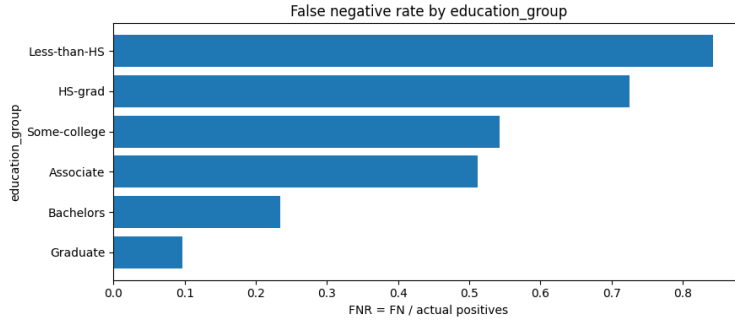


Figure 2: FNR across different education groups

4.2 Fairness Metrics Assessment

To evaluate fairness more directly, we use three group fairness metrics: statistical parity difference, equal opportunity difference, and false positive rate (FPR) gap. Statistical parity difference measures the gap in predicted positive rates across groups. Equal opportunity difference measures the gap in recall, or true positive rate, among individuals who actually earn $> 50K$. This metric is especially important in our setting because false negatives represent individuals in the $> 50K$ category who are incorrectly predicted as earning $\leq 50K$. The FPR gap measures whether individuals who actually earn $\leq 50K$ are incorrectly predicted as earning $> 50K$ at different rates across groups.

The statistical parity results show large differences in predicted positive rates. The sex-based gap is 16.7 percentage points: men are predicted as earning $> 50K$ at a rate of 24.7%, compared with only 8.0% for women. The race-based gap is 15.6 percentage points. The gaps are even larger for workclass, age group, marital status, and education: 41.5, 30.7, 37.9, and 68.6 percentage points, respectively. These results show that positive predictions are concentrated in certain social and economic groups.

The equal opportunity results show similar disparities. The sex-based gap is 10.3 percentage points, and the race-based gap is 23.8 percentage points. The gaps are larger for workclass, age group, marital status, and education, at 28.5, 43.0, 46.8, and 74.6 percentage points, respectively. The education gap is especially large: the model identifies graduate-degree holders who earn $> 50K$ at a high rate, but misses many $> 50K$ individuals in the Less-than-HS and HS-grad groups.

The FPR gaps show that the model also makes different types of errors across groups. The FPR gap is 6.2 percentage points for sex and 7.7 percentage points for race, but increases to 23.6 percentage points for workclass, 18.6 percentage points for marital status, and 34.9 percentage points for education. In general, groups that the model strongly associates with higher income, such as Graduate or Self-emp-inc, tend to have higher recall but also more false positives. By contrast, groups that the model weakly associates with higher income, such as Less-than-HS, HS-grad, Never-married, and younger individuals, receive fewer positive predictions and experience more false negatives.

Overall, these results suggest that the ADS does not satisfy statistical parity, equal opportunity, or false positive rate parity. The strongest disparities appear across education, marital status, age group, and workclass. Sex and race also show meaningful disparities, even though race was removed from the training features. This challenges the idea that fairness can be achieved simply by dropping a sensitive attribute, since other variables may still act as proxies and produce unequal outcomes.

Attribute	Statistical Parity Gap	Equal Opportunity Gap	FPR Gap
Sex	16.7	10.3	6.2
Race	15.6	23.8	7.7
Workclass	41.5	28.5	23.6
Age group	30.7	43.0	12.8
Marital status	37.9	46.8	18.6
Education group	68.6	74.6	34.9

Table 3: Fairness metric gaps across audited subgroup attributes

4.3 Intersectional Fairness Analysis

Single-attribute subgroup analysis can hide compounded disparities. To examine this issue, we evaluate intersectional groups defined by race $\times$ sex, race $\times$ education, and marital status $\times$ sex. These results show that some of the largest false negative rates appear only when multiple demographic or socio-economic attributes are considered together.

Among race $\times$ sex groups, Asian-Pac-Islander women have a false negative rate of 60.0%, Black women have a false negative rate of 52.6%, and White women have a false negative rate of 48.8%. These are all substantially higher than the false negative rate for White men, which is about 39.0%. This suggests that the model is less reliable for several female subgroups, especially when sex is analyzed together with race.

The race $\times$ education results show an even stronger pattern. White individuals with less than a high school education have a false negative rate of 83.7%, Black individuals with less than a high school education have a false negative rate of 80.0%, Asian-Pac-Islander individuals with only a high school degree have a false negative rate of 75.0%, and White high school graduates have a false negative rate of 72.7%. In contrast, White graduate-degree holders have a false negative rate of only 9.0%. This gap suggests that the model relies heavily on education-related signals and often fails to recognize high-income individuals who do not match the usual high-education profile.

A similar pattern appears for marital status $\times$ sex. Widowed women have a false negative rate of 83.3%, never-married men have a false negative rate of 80.9%, widowed men have a false negative rate of 80.0%, divorced women have a false negative rate of 77.5%, and never-married women have a false negative rate of 75.7%. These groups often have high overall accuracy, but that accuracy is misleading because the model frequently predicts the majority class while missing many actual $> 50K$ cases.

Overall, the intersectional analysis shows that single-attribute subgroup metrics can understate the severity of model disparities. The disparities become more pronounced when multiple demographic and socio-economic attributes are considered jointly.

4.4 Debunking “Fairness Through Unawareness” via Proxy Analysis

The original implementation removes the **race** variable before training the model. However, removing a sensitive attribute does not guarantee that the model is race-neutral, because other variables may still carry race-related information.

To test this, we compared the original setting, where **race** is excluded, with a sensitivity setting where **race** is included. This comparison is used only as an audit tool, not as a recommendation to include **race** in deployment. The purpose is to examine whether removing **race** actually reduces bias. Overall model performance stayed almost the same, with accuracy around 85%, and the major fairness gaps did not disappear. In the original no-race model, racial groups still had different selection rates, recall values, false positive rates, and false negative rates. For example, the race-based false negative rate gap was about 23.8 percentage points, meaning that some racial groups were much more likely to be incorrectly classified as earning $\leq 50K$ even when their true label was $> 50K$.

These results suggest that race-related information may still be indirectly encoded through other features. To investigate this possibility, we conducted a proxy analysis using Cramér’s V, which measures association between categorical variables. As shown in Table 4, **native.country** has the strongest association with **race** among the candidate proxy variables, followed by **sex**, **relationship**, **marital.status**, **occupation**, and **education**. This suggests that race-related information remains partially encoded in other features, meaning that these variables may function as proxies even after the **race** feature is removed.

Feature	Association with Race: Cramer’s V	Number of Unique Values
native.country	0.408352	42
sex	0.117451	2
relationship	0.097436	6
marital.status	0.083133	7
occupation	0.075103	14
education	0.071899	16
education.group	0.063633	6
workclass	0.052984	8
age.group	0.025317	6

Table 4: Cramer’s V association between **race** and potential proxy variables.

4.5 Imputation Sensitivity Audit

The original implementation replaces hidden missing values in **workclass** and **occupation**, recorded as “?”, with the most frequent categories: **Private** and **Prof-specialty**. We audit this preprocessing choice because mode imputation assumes that individuals with missing employment information are similar to the majority category, which may hide meaningful patterns of missingness.

Missingness in this dataset does not appear to be random. Test-set rows with at least one original “?” value had a much lower positive base rate than rows without missing values, 13.3% versus 25.1%. Although the baseline model achieved higher accuracy on rows with missing values, 90.4% versus 84.6%, this is partly because the missing-value group contains fewer positive cases. The lower precision for this group, 63.9% compared with 74.7%, suggests that positive predictions are less reliable for individuals with missing information.

We then compared three imputation strategies: the author’s original mode imputation, treating missing values as an explicit **Unknown** category, and deleting rows with missing values. Treating missingness as its own category slightly improved overall performance: accuracy increased from 85.0% to 85.3%, recall increased from 58.9% to 59.3%, and the false negative rate decreased from 41.1% to 40.7%. Row deletion produced a different trade-off: it reduced the dataset from 32,395 to 30,042 rows, lowered accuracy slightly to 84.8%, but improved recall to 61.2% and reduced the false negative rate to 38.8%. This suggests that row deletion can improve sensitivity to positive cases, but at the cost of excluding individuals with missing information.

Changing the imputation strategy did not remove the main fairness disparities. For **sex**, the equal opportunity gap stayed close to 10 percentage points across all three strategies: 10.3 under mode imputation, 9.7 under missing-as-category, and 10.0 under row deletion. For **race**, the gap was less stable, increasing from 23.8 percentage points under mode imputation to 32.4 under missing-as-category and 51.2 under row deletion. Because some race groups are small, these values should be interpreted cautiously, but they still show that race-related disparities persist across preprocessing choices.

Overall, mode imputation is not ideal because it makes the data appear complete while hiding patterned missingness. We recommend treating missing values as an explicit **Unknown** category, since it is more transparent and slightly improves performance without removing records.

Imputation Strategy	n	Accuracy	Recall/TPR	FNR	Sex EO Gap	Race EO Gap
Mode imputation	32,395	85.0%	58.9%	41.1%	10.3 pp	23.8 pp
Missing as category	32,395	85.3%	59.3%	40.7%	9.7 pp	32.4 pp
Row deletion	30,042	84.8%	61.2%	38.8%	10.0 pp	51.2 pp

Table 5: Performance and fairness sensitivity across imputation strategies

4.6 Auditing Age-Based Sample Selection Bias and Robustness

The original notebook applies an IQR-based age filter and keeps only rows with ages strictly between -2 and 78 . Since the lower bound is below the minimum possible age in the dataset, this preprocessing step only removes older individuals. As shown in Table 6, the filter excludes 166 rows, or about 0.51% of the full dataset. Although this is a small fraction of the data, the excluded rows are not random: they have a much higher mean age, a lower positive income rate, and a somewhat different demographic composition than the kept rows.

Group	n	Mean Age	Age Range	> 50K Rate	Male	White
Kept by age filter	32,395	38.4	17–77	24.1%	66.9%	85.4%
Excluded by age filter	166	83.1	78–90	18.1%	69.9%	89.8%

Table 6: Population differences between rows kept and excluded by the author’s age filter.

To test whether the age filter was necessary for model quality, we compared the author’s filtered pipeline with a sensitivity setting that kept all age values. As shown in Table 7, removing the age filter slightly improved overall performance. Accuracy increased from 85.0% to 85.6%, recall increased from 58.9% to 60.0%, and the false negative rate decreased from 41.1% to 40.0%. ROC-AUC also improved from 0.904 to 0.910. These changes are small, but they suggest that the age filter does not provide a clear technical benefit.

Setting	n	Accuracy	Recall/TPR	FNR	ROC-AUC
Author age filter	32,395	85.0%	58.9%	41.1%	0.904
No age filter	32,561	85.6%	60.0%	40.0%	0.910

Table 7: Performance sensitivity with and without the author’s age filter.

We also evaluated the trained baseline model on the 166 rows excluded by the author’s age filter as a robustness stress test. On this excluded older subgroup, the model achieved 81.3% accuracy and 80.0% recall, but its false positive rate was much higher at 18.4%, and its precision was only 49.0%. This indicates that the model does not completely fail on the excluded older individuals, but its error pattern changes substantially outside the filtered age range. Overall, the age-filtering step is difficult to justify: it removes a real older subpopulation, narrows the represented population, and does not improve model performance. A more responsible approach would be to keep these older individuals in the dataset and report age-group performance separately rather than treating them as outliers.

4.7 Transparency via Interpretable Coefficients

Because the baseline ADS uses Logistic Regression, we can inspect the model coefficients to understand which features most strongly push the prediction toward $> 50K$ or $\leq 50K$. Since the numerical features were scaled and the categorical features were one-hot encoded, the coefficients should be interpreted as effects in the processed feature space rather than as raw real-world causal effects. Still, they are useful for auditing the logic of the model.

The largest positive coefficient is `capital.gain`, with a coefficient of 2.28. This means that high capital gain strongly increases the model’s predicted probability of income > 50K. The next strongest positive features are `marital.status_Married-civ-spouse` (1.07), `education.num` (0.58), `sex_Male` (0.44), `hours.per.week` (0.40), and `age` (0.34). These results show that the model gives high-income predictions mostly to people with capital gains, higher education, longer working hours, older age, male sex, and married-civilian-spouse status.

The strongest negative coefficients also match the subgroup disparities found earlier. `education_Preschool` has a large negative coefficient of -0.57 . Other negative features include `relationship_Own-child` (-0.29), `occupation_Priv-house-serv` (-0.25), `workclass_Self-emp-not-inc` (-0.25), `workclass_Private` (-0.23), `occupation_Other-service` (-0.22), and `marital.status_Never-married` (-0.19). In other words, the model lowers the probability of > 50K for people who are younger dependents, never married, in certain service occupations, or in some workclass categories. This helps explain why groups such as younger individuals, never-married individuals, lower-education groups, and some occupation groups had high false negative rates in the subgroup analysis.

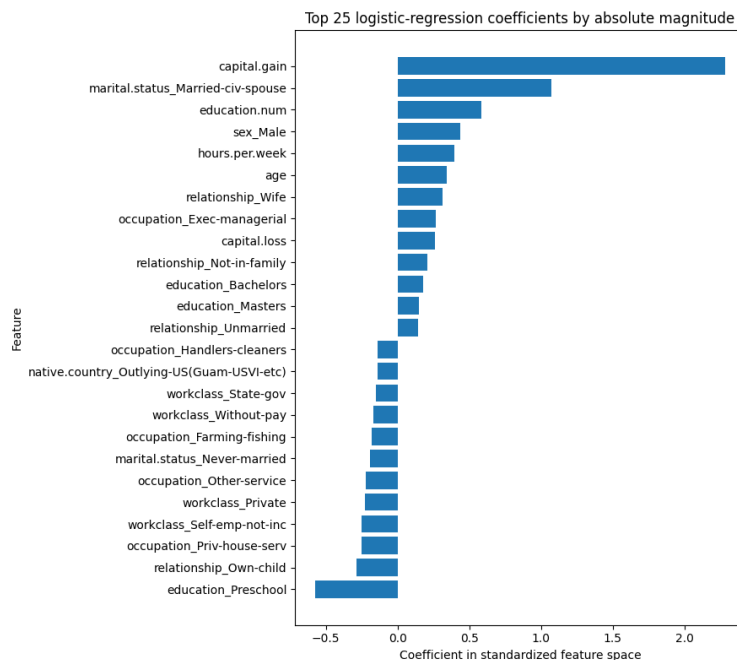


Figure 3: Logistic Regression coefficients

Overall, the interpretability analysis shows that the ADS is technically transparent in the sense that we can identify which features contribute most to its predictions. However, there is a trade-off between predictive performance and transparency. Simpler models, such as logistic regression, are easier to interpret, but they may achieve lower accuracy than more complex models. In this setting, the ADS is used for a socially consequential prediction task, so transparency should be treated as an important design goal rather than a secondary concern.

4.8 Conformal Prediction

In addition to accuracy and fairness metrics, we used conformal prediction to audit the uncertainty of the baseline model. Since this is a binary classification task, we used probability-based split conformal prediction. For each calibration example, we computed the nonconformity score

$$s_i = 1 - \hat{P}(Y = y_i \mid X = x_i),$$

where $\hat{P}(Y = y_i \mid X = x_i)$ is the model’s predicted probability for the true class. We then used the corrected 95% quantile of these scores as the threshold. For each test example, a label was included in the prediction

set if its predicted probability was at least $1 - \hat{q}$. Thus, the conformal set represents all labels that the model assigns sufficiently high probability to.

The overall empirical coverage on the test set was 94.8%, which is close to the intended level. The average prediction set size was 1.26, meaning that most predictions remained informative rather than always returning both labels. About 74.4% of test examples received a singleton prediction set, while 25.6% received the ambiguous two-label set. There were no empty prediction sets. In aggregate, the conformal procedure therefore achieved its intended goal: the true label was included in the prediction set for almost 95% of test examples.

However, subgroup results show that valid overall coverage does not guarantee equal reliability for every group. For **sex**, coverage was 96.0% for women and 94.3% for men. The male subgroup also had a larger average prediction set size, 1.33 compared with 1.10 for women, suggesting that the conformal layer more often treated male cases as uncertain. Coverage also varied across race groups. Black individuals had coverage of 97.0%, White individuals had coverage of 94.6%, Asian-Pac-Islander individuals had lower coverage at 92.4%, and the Other category had coverage of 94.0%. The Amer-Indian-Eskimo group had 100% coverage, but this group had only 65 test examples, so this number should not be over-interpreted.

The same pattern appears across age and education groups. The 45–54 and 55–64 age groups had lower coverage, 92.2% and 92.5%, respectively, even though they also had larger average prediction set sizes. Education produced one of the clearest coverage gaps: the Graduate group had only 90.1% coverage, and the Bachelors group had 93.6% coverage. By contrast, Less-than-HS and HS-grad were closer to or above the target, with 96.3% and 95.2% coverage, respectively. This suggests that stronger baseline classification performance does not necessarily imply better uncertainty calibration.

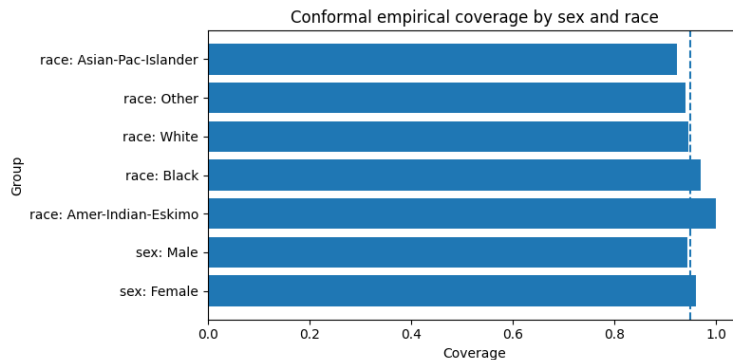


Figure 4: Subgroup conformal empirical coverage

Overall, conformal prediction provides a useful uncertainty layer on top of the classifier, and its average prediction set size remained reasonably small. However, standard split conformal prediction provides marginal coverage over the full distribution, but it does not guarantee equal coverage for every subgroup. For responsible deployment, we would not rely only on overall conformal validity. We would also report group-level coverage and consider group-conditional or fairness-aware conformal methods if this ADS were used in a real decision-support setting.

4.9 Privacy Risk Audit and Differential Privacy

The original implementation does not include a formal privacy-preserving mechanism. Although the Adult Census dataset is a public benchmark, it contains sensitive demographic and employment-related features such as age, education, marital status, occupation, race, sex, native country, and income. We therefore audit whether the data creates potential privacy risks.

We conducted a quasi-identifier uniqueness audit using age, education, marital status, occupation, race, sex, and native country. Out of 32,561 records, the median equivalence class size was only 3. In addition, 36.8% of records were unique under this quasi-identifier combination, 61.9% had an equivalence class size below 5, and 77.2% had an equivalence class size below 10. These results do not prove re-identification, but

they show that many records are highly specific. If an attacker knew a person’s demographic and employment profile, they might be able to narrow the person to a small set of rows and infer the sensitive income label.

This risk motivates data minimization and privacy protection. A more responsible deployment would justify each feature, generalize rare or identifying values, and avoid releasing row-level records. For example, exact age could be replaced with age bins, rare native-country values could be grouped into broader regions, and detailed occupation categories could be generalized. If model outputs, coefficients, or aggregate statistics were released publicly, differential privacy could provide a formal privacy guarantee by limiting the influence of any single record. However, this would introduce a privacy–utility trade-off, so different privacy budgets should be evaluated using accuracy, recall, false negative rate, and subgroup fairness gaps.

5 Summary

Overall, the Adult Census dataset is only partially appropriate for this ADS. It contains several features that are predictive of income, such as age, education, occupation, hours worked per week, and capital gains or losses. However, it also contains sensitive demographic and employment-related attributes, including race, sex, marital status, native country, and income. Our audits showed that missingness in employment-related features is not random, and that the author’s age filter removes a specific older subpopulation. Therefore, the dataset is useful as a benchmark, but it should not be treated as neutral, fully representative, or ready for real-world deployment without additional documentation and safeguards.

The implementation achieves reasonable aggregate accuracy, but it is not fully robust or fair. We evaluated accuracy, recall, false negative rate, precision, and average precision because this is an imbalanced binary classification task where accuracy alone can be misleading. We also used fairness metrics such as statistical parity, equal opportunity, false positive rate gaps, and false negative rate gaps because different stakeholders may care about different types of errors. Individuals affected by the model may care about whether they are incorrectly classified, policymakers may care about whether errors are distributed fairly across groups, and system designers may care about model stability. Our results show persistent sex- and race-based disparities, and these disparities remain across alternative imputation strategies. The conformal prediction audit also shows that overall uncertainty coverage can be close to 95% while still falling below target for some subgroups.

We would not be comfortable deploying this ADS in the public sector or in a high-stakes industry setting in its current form. Although the model may be useful for exploratory analysis, its preprocessing choices are not sufficiently justified, its subgroup performance is uneven, and its privacy protections are limited. Since income prediction is socially consequential, deployment would require more than good aggregate performance. It would require transparent preprocessing, subgroup-level reporting, privacy safeguards, and clear communication of uncertainty.

We recommend treating missing values as an explicit **Unknown** category, removing or better justifying the age filter, and reporting performance separately across sex, race, age, and education groups. We also recommend retaining sensitive attributes for internal fairness auditing while protecting them from unnecessary exposure. Rare quasi-identifier combinations should be generalized or suppressed before any data release, and differential privacy should be considered if model outputs, coefficients, or aggregate statistics are released publicly. Finally, if the ADS were used in practice, conformal prediction or another uncertainty method should be included, with both overall and subgroup-level coverage reported.

6 Project Contributions

- Project plan: Sifan Zhang & Hollan Yuan
- Code implementation: Hollan Yuan
- Report & slides: Sifan Zhang

ADS: <https://www.kaggle.com/code/sanjay7013/adult-census-income/notebook>
Dataset: <https://www.kaggle.com/datasets/uciml/adult-census-income>